DC MOTOR

A beginner-friendly guide for learners aged 18+

Learn what a DC motor is, how it works, its main parts, basic equations, applications, and safe practical experiments.

1. Learning goals

- Explain the meaning of a DC motor and what it does.
- Identify the main parts of a small brushed DC motor.
- Describe how magnetic fields and current produce rotation.
- Understand speed, torque, voltage, current, and back EMF at a basic level.
- Try a simple low-voltage experiment safely.

2. What is a DC motor?

A DC (direct-current) motor is an electrical machine that converts electrical energy into mechanical energy, usually as rotation. A battery or DC power supply provides electrical energy; the motor's shaft provides movement.

Simple idea: electrical energy in → rotating motion out.

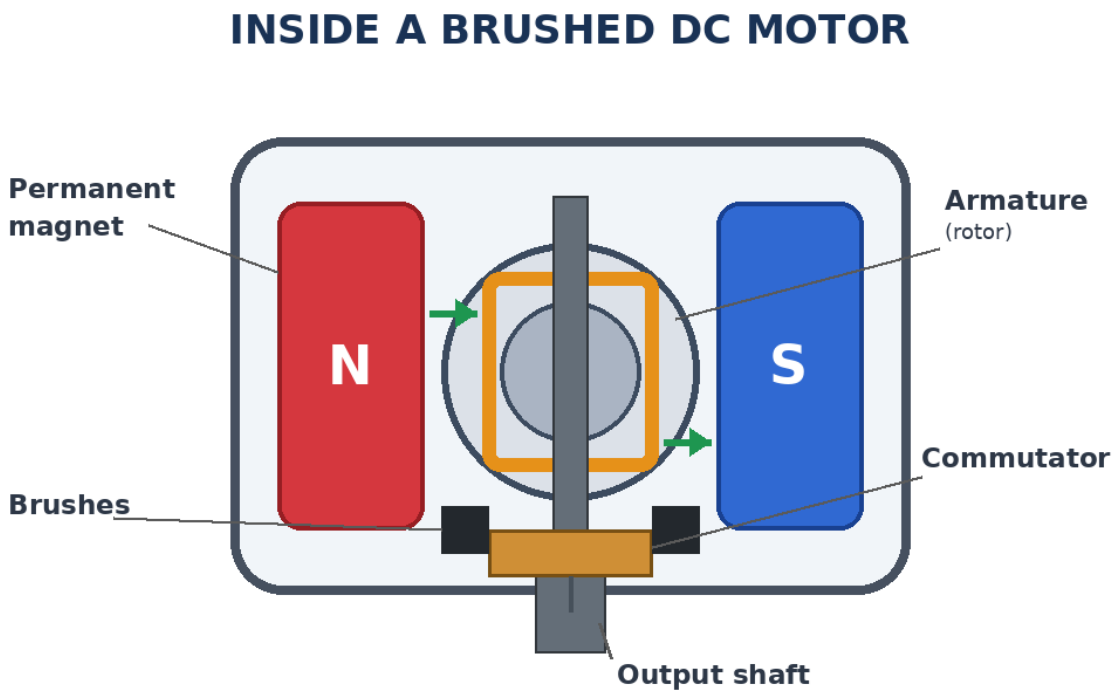


Figure 1. Simplified parts of a brushed DC motor. Real motor designs differ.

3. Main hardware parts

Part	What it is	What it does
Stator / field magnets	The stationary outer part	Creates the magnetic field.
Armature / rotor	The rotating part, often containing a coil	Carries current and experiences magnetic force.
Commutator	Segmented copper ring on the rotor	In a brushed motor, switches coil connections so torque continues in the same rotational direction.
Brushes	Usually carbon contacts	Transfer current to the rotating commutator.
Shaft	Metal rod connected to rotor	Transfers rotation to a wheel, fan, gear, or other load.
Housing / bearings	Outer frame and supports	Protects and supports the moving parts.

4. How does it work?

Step 1 — Supply voltage: A DC source pushes current through the motor's electrical circuit.

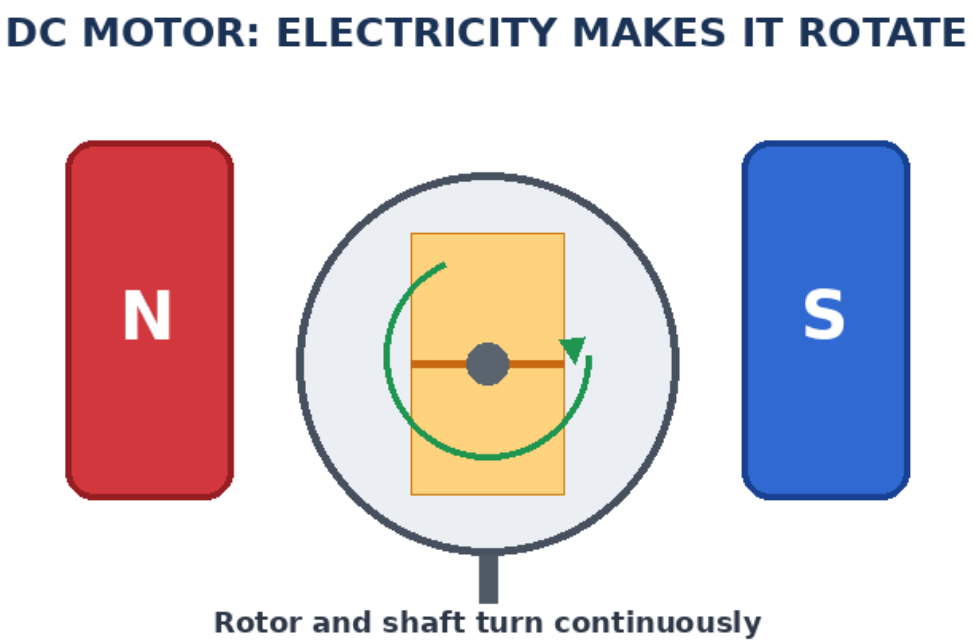
Step 2 — Magnetic interaction: A current-carrying conductor placed in a magnetic field experiences a force.

Step 3 — Torque is produced: Forces on different sides of the armature coil create a turning effect called torque.

Step 4 — Rotation continues: The commutator reverses the current in the coil at the appropriate point, helping the motor keep turning in the same direction.

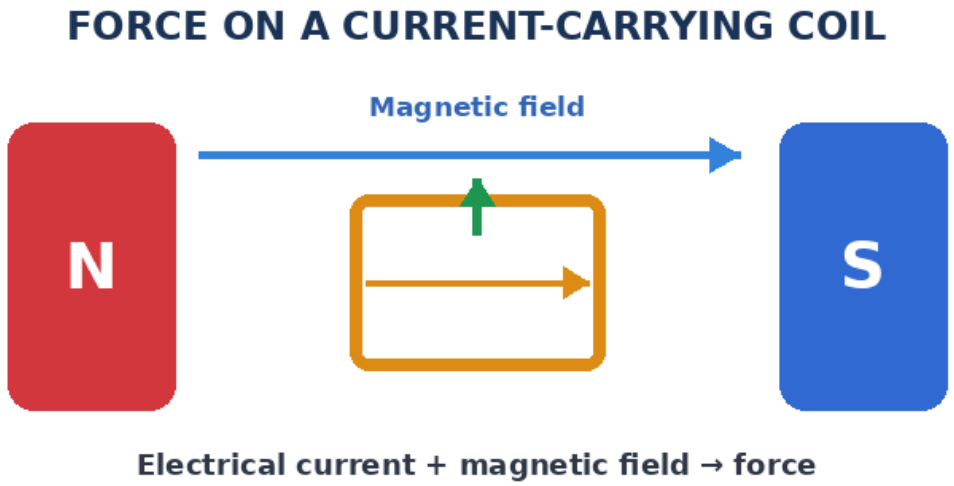
Step 5 — Useful work: The rotating shaft can drive a wheel, toy, pump, fan, or robot mechanism.

Animated illustration: open the accompanying file "dc_motor_rotation.gif" in a browser or image viewer to see a simplified rotor turning.



5. The physics in simple words

The motor effect is the force experienced by a current-carrying conductor in a magnetic field. The direction of the force depends on the direction of current and the magnetic field.



Animated concept illustration. The arrows are simplified and not a detailed electromagnetic simulation.

Fleming's left-hand rule

Hold the thumb, first finger, and middle finger of your LEFT hand at right angles to one another:

- First finger: magnetic Field (from North to South outside the magnet).
- Middle finger: conventional Current.
- Thumb: Force / motion of the conductor.

Memory tip: F-B-I — Force, magnetic field, current. Each direction is perpendicular to the other two.

6. Important quantities and beginner equations

Quantity	Meaning / formula
Voltage (V)	Electrical push supplied to the motor; measured in volts (V).
Current (I)	Flow of electric charge; measured in amperes (A).
Electrical input power	$P = V \times I$ (watts), for a DC input using the measured voltage and current.
Torque (τ)	Turning effect of the motor; measured in newton-metres (N·m).
Rotational speed	Often stated in revolutions per minute (rpm).
Back EMF	As the motor spins, it generates a voltage that opposes the supply. This helps limit current while running.

For an idealized motor, increasing the load often increases current and reduces speed. Actual behaviour depends on the motor, supply, controller, and load.

7. Types of DC motors

Type	How it is built	Common uses / note
Brushed DC motor	Uses brushes and a commutator	Simple toys, small mechanisms; brushes wear over time.
Brushless DC (BLDC) motor	Uses electronic switching instead of brushes	Drones, computer fans, e-bikes; requires a controller.
DC geared motor	DC motor with a gearbox	Robots and wheels; lower speed with higher output torque.

8. Where are DC motors used?

- Robotics and small mobile robots
- Battery-powered toys and models
- Automotive window and seat mechanisms
- Small pumps and portable tools
- Fans, actuators, and automation prototypes

9. Safe mini practical: make a motor spin

You will need: a small low-voltage DC motor rated for the supply, a matching battery pack or current-limited DC supply, insulated connecting wires, and (optional) a switch.

1. Read the motor's rated voltage and use a suitable supply. Do not guess the voltage.
2. With power disconnected, connect the motor terminals to the supply using insulated wires.
3. Secure the motor so it cannot roll or pull wires into the shaft.
4. Turn on the supply briefly and observe the shaft.
5. Disconnect power before changing wiring. Reverse the two motor leads to observe that a simple brushed DC motor usually reverses direction.
6. Never short-circuit a battery, touch exposed wires together, stall the motor for long, or connect a motor directly to mains electricity.

Safety note: Use low voltage and adult/teacher supervision where appropriate. A motor can heat up, and its shaft can catch hair, clothing, or fingers.

10. Troubleshooting basics

Symptom	Possible checks
Motor does not spin	Check supply voltage, battery charge, loose connections, switch, and whether the shaft is blocked.
Motor is weak or slow	Check battery condition, rated voltage, excessive mechanical load, and wire/contact quality.
Motor gets hot	Disconnect it; check for overload, stalled shaft, wrong voltage, or too much current.
Direction is wrong	For a brushed DC motor, power off and swap the two motor leads if the application allows it.

11. Check your understanding

1. What energy conversion takes place in a DC motor?

2. What is the role of the commutator in a brushed DC motor?

3. What does torque mean?

4. What does Fleming's left-hand rule help you find?

5. Name two devices that use DC motors.

6. Why should you disconnect the supply before changing the wiring?

Answer key

1. Electrical energy into mechanical energy.

2. It switches current in the armature coil at the right time to help maintain torque in the same direction.

3. The turning effect of a force.

4. The direction of force/motion for a current-carrying conductor in a magnetic field.

5. Examples: robot wheels, toys, pumps, portable tools, automotive mechanisms.

6. To reduce the risk of short circuits, unexpected movement, and damage.

12. Quick recap

A DC motor converts electrical energy into rotation. Current in the armature interacts with a magnetic field to create force and torque. In a brushed motor, brushes and the commutator help the rotor keep turning. Always use the correct low-voltage supply and safe wiring practices.

Keep learning: understand the parts → observe the motion → explain the physics → build safely.